

ENGR 415

C603: AIAA Student Competition Rocket Team (SCRT)

Roll Control Development

Project Scope and Research Statement

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Project Scope

The customer for this project is the American Institute of Aeronautics and Astronautics (AIAA) Oregon State Student Competition Rocket Team (SCRT) which designs and builds rockets for the AIAA competitions held for universities. A recurring challenge for SCRT rockets is the ability to maintain a stable attitude profile and roll control during boost. For this project, attitude profile describes the vehicle's orientation (roll, pitch, and yaw), and roll control refers to controlling the rotation about the rocket's longitudinal axis. The customer has also indicated that the current project state includes no completed trade studies, no prototypes, and a current rocket that does not include any features of roll control. This means that the roll-control project is effectively starting from scratch and must remain aligned with SCRT's integration needs and competition expectations (such as safety approvals, required design review deliverables, and flight readiness documentation) [1].

The focus of this project will be defining, assessing, and validating a practical roll-control solution for the SCRT team. The team will start with requirements by using the customer goals and competition/safety expectations to create measurable engineering requirements and constraints [1]-[3]. This will be followed by a trade study of candidate roll-control concepts applicable to SCRT's full-scale rockets, comparing available options such as trim tabs, fin flaps, canards, and fin cants against weighted criteria including authority, complexity, reliability, manufacturability, integration risk, and testability [4], [5]. Further, the project will also include simplified dynamics and aerodynamic control analyses to estimate the roll torque margins across flight and to identify key sensitivities [4]-[6]. Ultimately, the team will design and build a working sub-scale prototype to demonstrate the concepts in a controlled, testable configuration, supported by modeling, bench testing, and limited flight testing, resulting in documentation and deliverables that supports a path to the full-scale SCRT vehicle [1]-[3].

The following will define project success: (1) a trade study with documentation resulting in a justified concept; (2) a control analysis with documented assumptions and margins across flight; (3) a working sub-scale prototype that repeatedly actuates and demonstrates integrated roll-control operation within a test environment; and (4) a design that is ready to hand off to SCRT to scale this concept to the full-scale vehicle [1]. This project will not deliver a final, flight-ready roll-control system for the competition rocket, nor will it attempt a full guidance, and control solution beyond what is required to evaluate roll-control feasibility and demonstrate prototype operation.

Research and Previous Approaches

Roll-control and attitude control for fin-stabilized rockets are typically obtained through both active control and a selection of passive stabilization components. Passive means make use of aerodynamic and structural design choices [4]. Symmetry, fin sizing choice and mass balance increase available damping and reduce sensitivity to different aerodynamic disturbances. One

such example is spin stabilization due to fin cant which can help to average small asymmetries. However, there is limited ability to correct transient roll disturbance and dynamically adapt to flight conditions using passive means alone [4].

Active roll-control methods fall into aerodynamic actuation and internal torque generation. Aerodynamic actuation creates a roll torque by changing and redirecting the forces acting on the rocket's external surfaces, such as applying trim tabs or fin flaps, which create and alter lift forces that result in roll moments about the rocket's axis [4]. Aerodynamic actuation can be lightweight and mechanically straightforward, but their authority depends strongly on dynamic pressure and must account for both actuator limits and aerodynamic coupling [4]. This is relevant to the customer because they had previously noted tight trim-tab/fin packaging constraints and interest in linear actuation, which limits feasible implementations and favors concepts that are compact and buildable [1]. Additionally, front-mounted control surfaces such as canards can offer very strong control authority but can increase drag and introduce additional complexity that raises the verification requirements [5]. Internal torque generation approaches can include propulsion-based actuation, or other internal systems used in attitude control systems, but these approaches can lead to increased complexity, integration problems, and safety/compliance issues [6].

As an example of relevant prior research, the Portland State Aerospace Society (PSAS) has published techniques related to an active roll-control module utilizing small canard-style fins as seen in Figure 1. Their tests involve a fail-safe neutral behavior, short mechanical travel to minimize control gain, and mechanical linkage such that the surfaces work in concert primarily about the roll axis [7]. Additionally, PSAS leverages the insight that surfaces positioned close to the vehicle CG (center of gravity) minimize unintended steering in the case of imperfect alignment. They used a single high-authority actuator to simplify the roll-control system [7]. PSAS flight research and work further highlights that aerodynamic interference between forward surfaces and aft fins can produce unexpected behavior (including control reversal), reinforcing the need to acknowledge and assess coupling effects early in analysis and testing [7].

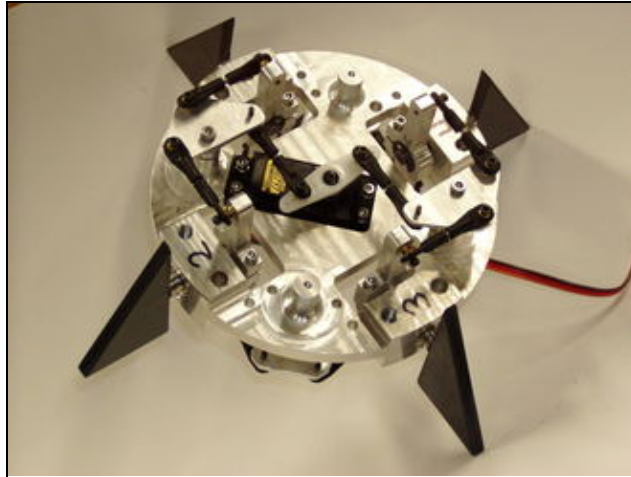


Figure 1. Active aerodynamic roll-control module developed by the Portland State Aerospace Society (PSAS), multiple control surfaces driven by a single actuator. Adapted from [7].

Building on these existing methods and research, this project's research will emphasize a concept screening and a selection process supported by modeling of control authority and sensitivities, and followed by prototype testing to confirm key assumptions and prove feasibility on a small-scale system.

Stakeholders

The primary stakeholder for this project is the customer, the AIAA Oregon State Student Competition Rocket Team (SCRT). Their requirements and expectations include implementing a roll-control system that can be justified through analysis, is compliant with the competition and safety expectations, and can be implemented in the full-scale SCRT rocket. External stakeholders include the competition organizers and the review personnel. They set requirements and expectations for evaluating the compliance and safety of designs, and their input can create constraints that can define concept selection and test planning. Additional project stakeholders include our technical advisors, and our managers, the MIME 497 instructors and assistants. They evaluate the project in terms of technical documentation, scope, discipline, research and engineering quality, deliverables, and they establish schedule constraints through assignments. Future members of SCRT are long-term project stakeholders. The value this project provides to SCRT as their customer does not end with the demonstration of a successful prototype. Future members of this club will rely on clear documentation, analysis assumptions, computer-aided design (CAD) files, test procedures, lessons learned, etc., to scale, replicate, and maintain our design in future years.

Constraints, Risks, and assumptions

This project is constrained by the customer's safety expectations and by competition rules requirements, which affect our concepts and documentation [1]-[3]. Physical integration is also a

major constraint due to limited packaging volume near the fin can and actuator mounting space; the customer noted trim-tab/fin space constraints and linear actuation packaging limits [1]. Schedule constraints limit the number of iterations possible, so designs must balance performance with buildability and testability [1]. Finally, resource and testing constraints (tool access, electronics support, computing availability, range access, and approvals) all limit the achievable accuracy of analysis and the number/type of tests that can be completed [1]. Safety reviews or administrative approvals may also delay access to test facilities, affecting schedule and the feasible number of iterations.

The primary technical risk would be inadequate roll-control authority levels for the expected flight profile, particularly in the presence of disturbances [4]. The aerodynamic/dynamics models may be too simple to account for the critical effects, which results in the risk that the prototype will behave poorly, compared to predictions [4], [5]. Integration risk may similarly be high, due to the potential impact of added hardware on structures, avionics or recovery systems. Schedule and compliance risks would also be present if part lead times, test availability or the need for documentation resulted in schedule impacts or limited the scope of testing [1].

This project also assumes the customer can provide baseline geometry and mass properties early enough to complete the trade study and analysis [1]. It also assumes at least one feasible validation pathway will be available, such as bench testing and possibly limited small-scale flight testing [1]. Lastly, it assumes that reduced modeling is a sufficient method for early down-select decisions and that coordination with other SCRT subteams will occur to define interfaces and avoid conflicts.

Conclusion

For the SCRT roll-control capstone project, there are currently no prototypes and no existing trade studies. The project is expected to develop from its requirements to a validated concept within a timeline. The scope of work in this document targets the highest near-term value to SCRT: a concept down-select with evidence, control authority assessment, and a functioning small-scale prototype to demonstrate feasibility and reveal integration challenges. The proposed research plan is to leverage existing roll-control mechanisms for similar rocket vehicles, adapt these mechanisms to the particular constraints of the SCRT rocket, and make design decisions consistent with competition judgment and review understanding [2], [3].

References

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